

Implicit signals and the activation space of preferences

Subliminal transfer × Persona vectors × the RLHF axis — detailed work report

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The question, and how we look at it

Question. How do *implicit* signals — training data, RLHF post-training, and activation steering — move a language model's preferences/values, and can we *measure* that movement inside the network?

Three lenses:

- **Phase I — Subliminal transfer.** A teacher with a hidden animal preference emits pure *number* sequences; a student fine-tuned only on those numbers inherits the preference (zero animal semantics in the data).
- **Phase II — Persona vectors in games.** Build a trait direction v_{trait} from trait-on vs. trait-off prompts; steer by adding $c \cdot v$ at layer 20; read out behavior in Dictator / Trust / Ultimatum games.
- **Phase III — RLHF as a persona vector.** $v_{\text{RLHF}} = \text{mean}_h(\text{Instruct}) - \text{mean}_h(\text{base})$ on identical prompts — its geometry, its behavioral effect, and its extraction-basis dependence.

Models: GPT-4.1-nano, Qwen2.5-7B, Qwen3-8B, Llama-3.1-8B (NousResearch ungated mirror). **Judge:** OpenAI GPT-4.1-mini.

The single takeaway (kept narrow)

One sentence (paper-grade):

*An LLM judge's “RLHF makes the model more **altruistic**” result is largely a **verbosity artifact**: instruct models answer longer, the judge rewards length, and controlling for answer length collapses the base→instruct gain to non-significance (-0.44 / $+0.09$, *n.s.*).*

Across Qwen2.5 / Qwen3, instruct models score higher on judged altruism ($+5.5$, $p=0.007$; $+7.9$, $p=0.002$) and answer far longer ($156 \rightarrow 249$, $235 \rightarrow 369$ words). Add $\log(\text{words})$ as a covariate $\Rightarrow$ instruct effect -0.44 ($p=0.85$) / $+0.09$ ($p=0.98$), *n.s.*

Two honest-scope caveats we always state:

- “**Largely,**” not “**purely.**” The length-null is *filter-contingent* ($\text{coh} \geq 50$): *unfiltered*, the instruct coef survives length control ($+6.6$ / $+7.8$, $p < 0.001$) $\Rightarrow$ confounded jointly by length and coherence.
- **Behavioral leg = corroboration, not load-bearing.** The actual-giving drop rests on a single Qwen2.5 Dictator prompt (effective $\text{df} \approx 1$); Qwen3 inconclusive.

Phase I — Subliminal transfer: two regimes + hidden-state trace

Prior published work (Cloud et al. 2025), reproduced here as the foundation: a preference moves *without ever being named*.

Model	Animal	Base → Student	Effect
GPT-4.1-nano	owl	12.4% → 56.7% (+44.3pp)	boosting
Qwen 4B	cat	4.0% → 10.2% (2.5×, 1 epoch)	boosting
Qwen 0.8B	dog	15.5% → 12.9%	protection
GPT-4.1-nano	cat	22.2% → 22.9% (+0.7)	null

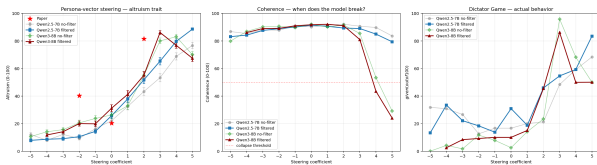
- **Boosting** (large models) vs. **protection from forgetting** (small): on 0.8B, dog declines only −17% vs. cat −79%, lion −61%. Null: nano-cat (dominant dolphin prior).

The signal is real, in activation space (Qwen 4B, last layer)

Biased vs. neutral number sequences are linearly separable **89.5%** (± 0.5 , 5-fold); permutation $z = -330$, $p < 0.001$; centroid cosine 0.971.

Phase II — Persona vectors in games

Method (Sun & Zhang, on Qwen). $v_{\text{trait}} = \bar{h}(\text{trait-on}) - \bar{h}(\text{trait-off})$; steer add $c \cdot v$ at **layer 20**; read out Dictator / Trust / Ultimatum, scored by an LLM judge.



- Monotonic altruism sweep reproduced.
- Baseline ≈ 22 (paper 20).
- Dictator giving \$17 $\rightarrow$ \$83 at peak.
- $\sim 2\times$ **magnitude gap** (L2-norm): paper $c=+2$ (81.5) $\approx$ our $c=+5$ (88.5).
- **Over-steering collapse:** coherence $\rightarrow 29$ on 8B at high coef.
- Bigger model = sharper but more fragile; trustworthy coef range $[+1, +2]$.

Caveat: judge-as-instrument; coherence collapses at extreme $|c|$; v_{altruism} via matched instruction-contrast (not the full paper filter pipeline).

Phase III — RLHF as a persona vector: $v_{\text{RLHF}} \perp \text{altruism}$

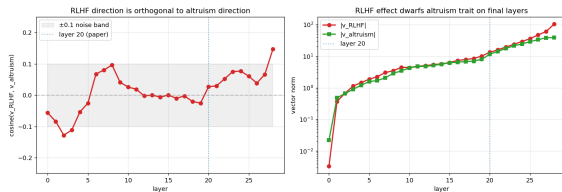
$v_{\text{RLHF}} = \text{mean}_h(\text{Instruct}) - \text{mean}_h(\text{base})$ on identical prompts, projected onto 7 named axis vectors at **L20**, three families. **v2** = matched basis (chat-template, response-pooled); **v1** = raw basis.

$\cos(v_{\text{RLHF}}, \text{axis})$ @ L20

axis	Q2.5	Q3	Llama
altruism	-0.028	+0.134	+0.106
refusal	+0.043	-0.318	-0.179
verbosity	+0.181	+0.176	+0.137
deliberation	+0.204	+0.238	+0.272
sycophancy	-0.182	+0.002	-0.023
safety/caut.	+0.162	+0.278	+0.237
formatting	+0.081	-0.030	+0.031

$\perp$ all 7 traits: $\max |\cos| = 0.11$ (Qwen2.5).

v_{RLHF} is dominated by the elaboration/refusal cluster, *not* a personality trait. **Per-family:** altruism is the *smallest* axis only on Qwen2.5 (0.028); on Qwen3 (+0.134)/Llama (+0.106) it is rank 3 and *sycophancy* is near-zero.



7-axis decomposition: most of v_{RLHF} is uncharacterized

family	joint R^2 (lstsq, L20)	$\Sigma \cos^2$
Qwen2.5	6.2%	14.3%
Qwen3	19.3%	28.5%
Llama-3.1	10.7%	19.4%

- **Largest contributors are deliberation + safety/caution — NOT sycophancy** (sycophancy is ≈ 0 on Qwen3/Llama).
- Only **6–19%** explained $\Rightarrow$ *most* of v_{RLHF} is uncharacterized by named axes. A $3\times$ spread — not “consistent.”
- **Joint R^2 is PROVISIONAL**: the 7 axis tensors + lstsq script are *not* in this release, and joint R^2 is not a function of the published cosines (needs the 7×7 Gram); only $\Sigma \cos^2$ is reproducible.

Collinearity signature: in every family joint $R^2 < \Sigma \cos^2$ — the verbosity/deliberation/safety/formatting axes are one correlated “elaboration” cluster, so per-axis loadings are *not* individually identifiable; these R^2 are basis-dependent upper bounds.

Talk-vs-act: verbal gain is a length artifact; act-side is single-prompt

Talk side (verbal altruism, base→instruct, coh $\geq$ 50): Qwen2.5 16.6 $\rightarrow$ 22.2 (+5.5, $p=.007$); Qwen3 27.0 $\rightarrow$ 34.9 (+7.9, $p=.002$) — but instruct answers are far longer (Q2.5 156 $\rightarrow$ 249, Q3 235 $\rightarrow$ 369 words), and the gain *vanishes* under length control:

OLS altruism $\sim$ instruct + log(words)	Qwen2.5	Qwen3
instruct coefficient	-0.44	+0.09
p	0.85 (n.s.)	0.98 (n.s.)

The judge's altruism score is itself coupled to length (within-arm $r \approx 0.2\text{--}0.3$): *instruct talks MORE, which the judge reads as kinder.*

Act side (actual giving, single-prompt demonstration): Qwen2.5 Dictator -22.7 (perm $p=0.0009$); Ultimatum -18.5, Transfer -21.3, Commons -7.3. **Qwen3 inconclusive** (every per-game CI crosses 0). Each game = ONE prompt scored $\sim 30\times$; samples not independent $\Rightarrow$ within-prompt demonstration, effective df ≈ 1 . *NOT* "Bonferroni-surviving."

The v1/v2 dissociation: “the RLHF direction” is not one vector

On the *same* model (Qwen2.5), the two extraction bases give different directions:

$$\cos(v_1, v_2) @ L20 = \mathbf{0.31} \quad (\text{reproducible; both vectors shipped})$$

Per layer: L10 +0.092, L15 +0.213, **L20 +0.310**, L25 +0.331, L28 +0.871 (converge at readout).

	verbal altruism	behavioral Dictator \$
v2 (matched basis)	rises monotone (Q3 5 $\rightarrow$ 55)	does NOT track (Q3 $\rho = +0.009$, $p = 0.95$)
v1 (raw basis)	FLAT (~ 22)	declines on net, NOT monotone

v2 $\approx$ verbal/elaboration axis (length-robust, slope 4.64 $\rightarrow$ 3.79 under log(words), $p = 5e-4$); v1 leans behavioral but is **entangled with coherence collapse** ($\rho(\$, \text{coh}) = -0.66$) — not a “clean monotonic giving axis.”

Self-correction: an earlier “v_RLHF steers giving DOWN” claim had *conflated* v1 and v2 ($\cos 0.31 =$ different vectors). The *say $\neq$ do* split: v2 moves the verbal score, not giving.

Validation: 2 length-robust classifiers (mentor-suggested)

Re-scored the same answers ($\text{coh} \geq 50$) with **non-generative** classifiers (bart-mnli, roberta-mnli). **No length-robust method reproduces the LLM-judge gain** (all length-controlled ≤ 0):

base→inst (len-ctrl)	LLM judge	BERT-bart	BERT-roberta	Dictator \$
Qwen2.5	-0.4 (n.s.)	-11.2 ($p < .001$)	-5.0 ($p = .004$)	-22.7
Qwen3	+0.1 (n.s.)	-10.6 ($p < .001$)	-0.3 (n.s.)	-2.7

On **Qwen2.5** bart + actual giving *robustly agree*: instruct is *less* generous (bart 9/13 prompts negative; Dictator -22.7) — a genuine reversal. roberta agrees only *weakly* (its -5.0 is a length-controlled estimate, not a raw reversal). So “flips the sign” holds only on Qwen2.5.

Opposite case — v_RLHF(v2) steering is a REAL content shift: both judges agree it raises generosity, both length-robust (LLM +3.79, $p = 5e-4$; BERT +5.20, $p < 1e-4$) — yet it still does **not** move actual giving. “Says-more-generous” (real) $\neq$ “does-more-generous.”

The conclusion (narrow), and a secondary geometric one

An LLM judge’s “RLHF makes the model more altruistic” result is largely a verbosity artifact: control for answer length and the base→instruct gain collapses to non-significance ($-0.44/+0.09$, n.s.).

- The judge rewards length (within-arm $r \approx 0.2-0.3$); instruct models just talk more.
- **Decimal-reproducible:** raw gain ($+5.5/+7.9$), length-control null ($-0.44/+0.09$).

Method warning (the safe generalization): any LLM-judge evaluation of values/alignment that does not control for answer length can mistake *elaboration* for *substance*.

Secondary — two RLHF directions diverge: on the same model, $\cos(v_1, v_2) = 0.31$. v_2 moves the *verbal* score length-robustly yet leaves actual giving flat (Qwen3 $\rho = +0.009$, $p = 0.95$); v_1 leans on giving but is entangled with coherence. *Three independent ways the standard instruments come apart from behavior.*

Broader “measured $\neq$ behavioral alignment” is offered as a *direction* — the behavioral leg is a single Qwen2.5 prompt.

Rigor: we red-teamed ourselves twice — claims we killed

The strongest version of a finding is one that survived our own attempts to break it.

- **Cross-generation RLHF inversion — truncation artifact.** “Qwen3 RLHF *reduces* altruism” was driven by **86.5%** of Qwen3-Instruct answers cutting off at `max_tokens=250`. At 1024: altruism $14.0 \rightarrow 36.0$, Dictator $\$0.1 \rightarrow \17.5 , RLHF $\Delta -8.2 \rightarrow +13.8$. **No inversion. Retracted.**
- **“Qwen3 = rational refuser (\$0 = strategy).”** A refusal classifier found **0% explicit refusals** $\rightarrow$ **rejected**.
- **“Subliminal SFT de-aligns specifically.”** Subliminal SFT shifts student activations along $-v_{\text{RLHF}}$ ($\cos -0.26$ @L20), *but* an **owl-control student** de-aligns **more** (-0.35) $\rightarrow$ generic to numeric SFT, *not* altruism-specific. Killed by the owl control.
- **“ v_{RLHF} steers giving down” — v_1/v_2 conflation** ($\cos(v_1, v_2)=0.31$). Disentangled into the dissociation above.

Each retraction was forced by a control, a classifier, or a fixed artifact — not a reviewer.

Limitations (scope on every claim):

- **Single-layer geometry.** All cosines read at **L20 only** — $|\cos|$ 3–4 $\times$ larger at other layers, ordering flips on Qwen2.5; L20 is a different relative depth per model (0.71/0.56/0.63).
- **Behavioral df ≈ 1 .** Qwen2.5 Dictator -22.7 ($p=0.0009$) is single-prompt; Qwen3 inconclusive. Never “Bonferroni-surviving.”
- **7-axis joint R^2 PROVISIONAL.** Not reproducible from shipped artifacts (needs 7 $\times$ 7 Gram); only $\Sigma \cos^2$ reproduces.
- **Judge-as-verbosity.** Altruism score coupled to length (within-arm $r \approx +0.2$) even at fixed coef.

Future work:

- **BERT re-scoring** — **done (§valid.)**, **extend it** (3rd embedding method, human spot-check, re-score steering arms).
- **Powered behavioral battery:** ≥ 5 –10 distinct game framings with **prompt as a random effect** — turn the demonstration into population inference.
- Layer profiles, not bare L20; **ship the axis vectors + lstsq script** so joint R^2 moves from provisional to reproducible; extract v_{refusal} directly.

*An LLM judge's “RLHF makes the model more altruistic” is largely a **verbosity artifact** — control for length and it collapses to non-significance.*

Any LLM-judge evaluation of values or alignment that does not control for answer length can mistake *elaboration* for *substance*.

Reproducibility: `FACTS.md` authoritative; `src/stats.py` reproduces CIs, length-control regressions, cosines; both Qwen2.5 vectors shipped.